Learning to Fit Shapes to Neural Geometry: A Synthetic-Only Shape-Type and Parameter Regressor for Top-3 PCA Activation Clouds

Libor "Burny" Burian

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Abstract

Interpretability research repeatedly reports low-dimensional shapes in neural activations: circles for weekdays and months, helices for positions and counts, a paraboloid for colors, lines for temperature and age, polytopes for features in superposition. These shapes are usually identified and fitted by hand. We (i) compile a cited catalog of empirically reported neural geometries, (ii) train a permutation-invariant network *only on synthetic shapes* that maps a 3D point cloud (the top-3 principal components of activations) to one of 13 shape families and to a canonical, unique parameter vector, using a shape-to-shape Chamfer loss against the ground-truth shape plus a point-to-shape residual, and (iii) evaluate it on a preregistered synthetic test set and on real activations of GPT-2, Llama-3.1-8B, a grokked modular-addition transformer and toy models of superposition, against a classical fit-every-family baseline with geometric-AIC model selection, a feature-based classifier, trivial baselines, random controls and ablations. On 39000 held-out synthetic clouds the fitter identifies the family in 87.9% of cases; on a shared subset it reaches 87.4% against 52.2% for the classical pipeline and 64.5% for the feature classifier, with a lower median Chamfer distance to the ground-truth shape (preregistered H1 and H2 supported). On real activations its coarse shape class agrees with published descriptions on 7/12 preregistered datasets, which is not above a permutation null (H3 not supported). Failure analysis traces this to curves outside the vocabulary, many discrete groups and low-variance projections. Controls show that smooth ordinal processes are often labelled helix, so helix claims need a held-out test against the PCA ringing artifact.

1 Introduction

A growing body of work shows that many concepts in language models are represented not by single directions but by low-dimensional manifolds [Engels et al., 2025, Bhalla et al., 2026, Gurnee et al., 2025]. The common workflow is: collect activations for a set of inputs that vary along one concept, project onto the top principal components, look at the 2D/3D scatter, and describe the shape ("a circle", "a helix"), often followed by a hand-designed parametric fit [Kantamneni and Tegmark, 2025]. This is slow, subjective, and vulnerable to a known artifact: principal component analysis (PCA) of *any* smoothly varying ordinal variable yields Fourier-like components and therefore an arch or a helix-like curve in the top components even when nothing "curved" is represented [Novembre and Stephens, 2008, Diaconis et al., 2008, Antognini and Sohl-Dickstein, 2018, Shinn, 2023].

We ask whether the description step can be automated: given the top-3 PCA coordinates of a set of activations, predict which shape family it is and the parameters that uniquely specify the shape, such that the shape passes as close as possible to the points. Our contributions:

1. A catalog of empirically reported neural geometries with model, layer, method, whether the shape is visible in the top-3 PCs, and verified sources (Sec. 2).
2. A 13-family shape vocabulary covering the catalog, with differentiable samplers, label margins between nested families, and canonical forms that make the output parameters unique (Sec. 3), including a family for the PCA "ringing" confounder.
3. A Set Transformer fitter trained only on on-the-fly synthetic shapes with domain randomization, a shape-to-shape Chamfer loss, and test-time refinement (Sec. 4).

Table 1: The 13 shape families. Local shapes are posed by a center c and a frame $R = (e_1, e_2, e_3)$ unless stated. Canonical forms make the reported parameters unique.

Family	Local shape	Canonical unique parameters	Anchor
blob	$\mathrm{diag}(\sigma_1, \sigma_2, \sigma_3)z$, $z \sim \mathcal{N}(0, I)$	center, axes (signs fixed), $\sigma_1 \geq \sigma_2 \geq \sigma_3$	null class
clusters	$k \in \{2..6\}$ centroids + spread	centroids sorted lexicographically, spread	true/false [Marks and Tegmark, 2024]
polytope	regular template (6 subtypes)	subtype, center, circumradius, sorted vertices	superposition [Elhage et al., 2022]
line	tLe_1, $t \in [-1, 1]$	endpoints	temperature, age [Bhalla et al., 2026]
arc	$r(\cos t\, e_1 + \sin t\, e_2)$, $\lvert t\rvert \leq \Delta/2$	center, normal, r, extent (closed if $\Delta \approx 2\pi$)	days, months [Engels et al., 2025]
ellipse	$a \cos t\, e_1 + b \sin t\, e_2$	center, normal, major axis, $a \geq b$	distorted circle
helix	$r \cos t\, e_1 + r \sin t\, e_2 + \frac{pt}{2\pi} e_3$	axis, r, signed pitch p, turns, phase	GPT-2 W_{pos} [Yedidia, 2023]
ringing	$\sum_{k=1}^{3} a_k \cos(k\pi s) e_k$	$a_1, a_2 > 0$, a_3, frame	PCA artifact [Novembre and Stephens, 2008]
disc	$\rho(a \cos t\, e_1 + b \sin t\, e_2)$	center, normal, $a \geq b$	pizza [Zhong et al., 2023]
sphere	cap of half-angle α around e_3	center, r, pole, α	cities [Tiblias et al., 2025]
paraboloid	$z = \frac{1}{2}(\kappa_1 x^2 + \kappa_2 y^2)$, $\lvert\kappa_2\rvert \leq \kappa_1$	frame, κ_1, κ_2, patch radius	colors [Bhalla et al., 2026]
cone	frustum $r(h) = r_0 + gh$ around e_3	axis, bottom/top radius, height	anisotropy [Ethayarajh, 2019]
torus	$((R + r\cos v)\cos u, (R + r\cos v)\sin u, r \sin v)$	center, axis, R, r	grid cells [Gardner et al., 2022]

4. A preregistered evaluation (hypotheses H1–H5) on synthetic and real activations with baselines, controls, ablations, bootstrap confidence intervals and multiple-testing correction (Sec. 5–6); both positive and negative results are reported.

2 Background and related work

Empirical neural geometries. Circles for days of the week and months were found with sparse-autoencoder clusters and PCA in GPT-2, Mistral-7B and Llama-3-8B [Engels et al., 2025]; GPT-2's learned positional embeddings form a helix in their top-3 PCs [Yedidia, 2023]; the character count since the last line break lives on a rippled helix-like curve in Claude 3.5 Haiku [Gurnee et al., 2025]; numbers form a generalized helix [Kantamneni and Tegmark, 2025]; Llama-3.1-8B layer 19 shows a paraboloid for colors, a helix for years, a circle for days and lines for temperature and age [Bhalla et al., 2026]; grokked modular-addition transformers place residues on circles ("clock") or discs ("pizza") [Nanda et al., 2023, Zhong et al., 2023]; features in superposition form regular polytopes [Elhage et al., 2022]; true and false statements separate into clusters [Marks and Tegmark, 2024]; supervised MDS recovers circles, lines and clusters for temporal concepts [Tiblias et al., 2025]; in-context graphs reorganize into rings and grids [Park et al., 2025]; emotions lie on a circle [Sun et al., 2026]. Outside LLMs, grid cells form a torus [Gardner et al., 2022], head-direction cells a ring [Chaudhuri et al., 2019], natural image patches a Klein bottle [Carlsson et al., 2008]. Appendix A (generated from `catalog.yaml`) lists all entries.

Primitive fitting. Deep primitive fitting on CAD point clouds predicts per-point memberships and fits planes, spheres, cylinders and cones with differentiable least squares [Li et al., 2019], B-spline patches [Sharma et al., 2020] or regressed parameters [Yan et al., 2021]; the classical alternative is RANSAC [Schnabel et al., 2007] with information-criterion model selection [Kanatani, 2004]. These methods target large CAD clouds with several primitives; neural-geometry clouds have one shape, few points (7 weekdays), high noise, families absent from CAD vocabularies (helix, torus, PCA ringing), and come from a PCA projection of a much higher-dimensional space. To our knowledge, no prior work learns shape-type classification plus parameter regression for activation point clouds; the closest is hypothesis-driven supervised MDS, which needs concept labels and has no free shape parameters [Tiblias et al., 2025].

3 Shape vocabulary

Each family f maps unconstrained raw parameters θ to a shape $S_f(\theta) \subset \mathbb{R}^3$, posed by a center c and a rotation $R \in SO(3)$ predicted with the continuous 6D representation [Zhou et al., 2019]. Table 1 lists the families with their parameters. Each family provides (i) a differentiable dense sampler on a fixed grid with length/area weights, (ii) label margins so that a generated instance is not (nearly) a member of a simpler family at the sample's noise level (e.g. an arc's sagitta must exceed $\max(0.1\,\mathrm{chord}, 3\sigma)$), and (iii) a canonicalization map that selects a unique representative among parameterizations of the same point set (axis signs fixed by the sign of the largest component, rotation about symmetry axes fixed by projecting the first PCA axis, point sets sorted lexicographically). The *ringing* family $c + \sum_{k=1}^{3} a_k \cos(k\pi s) e_k$ is the shape PCA produces for smooth ordinal data [Novembre and Stephens, 2008, Antognini and Sohl-Dickstein, 2018], so a helix claim must beat it.

4 Method

Preprocessing. Synthetic and real clouds pass through the same map: center, project onto the top-3 principal directions, fix axis signs by the sign of the third moment, force a right-handed frame (3D inputs), and divide by the RMS radius. No whitening is applied, so aspect ratios are preserved.

Synthetic data. Each training step draws fresh shapes for all 13 families from a dedicated seed stream (test and validation streams are disjoint by construction). Nuisances implement domain randomization [Tobin et al., 2017]: $N \in [5, 1024]$ log-uniform, relative noise $\sigma \in [0.003, 0.15]$ log-uniform (10% noise-free), anisotropic noise, non-uniform sampling densities, up to 10% outliers, and, for 25% of samples, embedding into $\mathbb{R}^D$ ($D \in \{16, 64, 256\}$) with noise in all D dimensions followed by PCA back to 3D, which reproduces the real pipeline including projection effects and mirrored frames.

Model. A Set Transformer encoder [Lee et al., 2019] (two ISAB blocks with 16 inducing points, PMA pooling, $d = 128$, 4 heads, key-padding masks) maps per-point features (xyz and $\log N$) to an embedding; a classification head outputs logits over the 13 families and one MLP head per family outputs its raw parameters.

Loss. For a sample of family y with ground-truth shape S^* and input points X,

$$\mathcal{L} = \mathrm{CE}(\ell, y) + \mathrm{CD}\big(S_y(\hat{\theta}_y), S^*\big) + 0.5\,\mathrm{Res}\big(X \to S_y(\hat{\theta}_y)\big) + \lambda(t)\,\mathrm{Aux}, \tag{1}$$

where CD is the symmetric, length/area-weighted, unsquared Chamfer distance between dense samples of the predicted and of the noise-free ground-truth shape (restricted to its true extent), Res is a Huber point-to-shape residual, and Aux is an annealed axis/scale term $1 - (\hat{a} \cdot a^*)^2 + |\log \hat{s} - \log s^*|$. Because CD compares point sets it is invariant to every parameter symmetry, so no canonicalization or matching is needed during training. Only the ground-truth family's head receives geometric losses (teacher forcing, as in Li et al., 2019).

Inference. The top-3 families are each refined for 300 Adam steps on the two-sided objective $\frac{1}{N}\sum_i d(x_i, S) + 0.25\sum_j w_j d(s_j, X)$, initialized by the network (amortized inference followed by optimization, cf. Kolotouros et al., 2019); the second term prevents degenerate inflation of radii and extents. The family is selected by the temperature-calibrated classifier [Guo et al., 2017]; geometric AIC [Kanatani, 2004] and a hybrid rule are ablations. The output is mapped back to PCA coordinates and canonicalized.

5 Experimental setup

Preregistration. Hypotheses, metrics, analysis plan and falsification criteria were committed before training (`reports/preregistration.md`); deviations are logged in `reports/deviations.md`.

Synthetic test set. 250 samples per cell of 13 families × 4 point-count bins × 3 noise bins (39,000 clouds) from the `test` stream. Baselines run on a fixed subset of 20 samples per family and cell; all paired comparisons use that subset.

Baselines. B1: classical multi-start fit of every family with the same two-sided objective (inits from the PCA frame and its axis permutations, k-means for clusters, every polytope subtype) and geometric-AIC selection. B2: hand-crafted features (global and local PCA eigenvalue ratios [Weinmann et al., 2015], persistent-homology H0/H1 lifetimes via ripser [Tralie et al., 2018], radial statistics) with gradient-boosted trees, then B1's fit of the predicted family. B3: always blob, always circle, random shape.

Ablations. A1 parameter-space loss, A2 no residual term, A3 no refinement, A4 DeepSets encoder [Zaheer et al., 2017], A5 selection rule, A6 no $\mathbb{R}^D$ augmentation; A1/A2/A4/A6 are compared with A0, the unchanged method trained with the same shorter budget.

Table 2: Synthetic test set. Fine accuracy is the exact family; coarse accuracy is the shape dimension (point set, curve, surface, blob). Chamfer is the median symmetric Chamfer distance between the selected, refined shape and the noise-free ground-truth shape. Baselines and the "baseline subset" row use the same 20 clouds per family and cell. 95% percentile-bootstrap CIs. The full table (superset-tolerant accuracy, residual, axis error) is in `reports/exp01_synthetic.md`.

System	n	Fine accuracy [95% CI]	Macro-F1 [95% CI]	Coarse accuracy	Median Chamfer to GT [95% CI]
Method (network + refinement)	39000	87.9% [87.6, 88.2]	0.879 [0.876, 0.882]	92.7%	0.156 [0.154, 0.158]
Method, baseline subset	3120	87.4% [86.2, 88.5]	0.873 [0.862, 0.884]	92.3%	0.154 [0.147, 0.161]
A3: no refinement	39000	87.9% [87.6, 88.2]	0.879 [0.876, 0.882]	92.7%	0.159 [0.157, 0.162]
A5: G-AIC selection	39000	64.6% [64.1, 65.1]	0.627 [0.623, 0.632]	77.5%	0.182 [0.179, 0.185]
A5: hybrid selection	39000	85.6% [85.2, 85.9]	0.858 [0.854, 0.861]	91.1%	0.157 [0.155, 0.159]
B1: classical fit + G-AIC	3120	52.2% [50.6, 54.0]	0.499 [0.484, 0.515]	70.7%	0.209 [0.200, 0.221]
B2: features + boosting	3120	64.5% [62.8, 66.2]	0.645 [0.629, 0.662]	85.1%	0.237 [0.221, 0.255]
B3a: always blob	3120	7.7% [6.8, 8.7]	0.011 [0.010, 0.012]	7.7%	0.543 [0.536, 0.550]
B3b: always circle	3120	7.7% [6.8, 8.6]	0.011 [0.010, 0.012]	38.5%	0.545 [0.536, 0.555]
B3c: random shape	3120	7.9% [7.1, 8.8]	0.079 [0.070, 0.088]	32.9%	1.116 [1.100, 1.137]

Real activations. R1 GPT-2 positional embeddings [Radford et al., 2019, Yedidia, 2023]; R2 embeddings of a grokked 1-layer modular-addition transformer [Nanda et al., 2023]; R3 toy models of superposition with 3 hidden dimensions [Elhage et al., 2022]; R4 Llama-3.1-8B layer-5 mean activations per character count since the last newline [Gurnee et al., 2025]; R5 weekday and month tokens in the templates of Engels et al. [2025] (layer 5); R6–R8 years, colors, temperature and age with the templates of Bhalla et al. [2026] (layer 19, last token); R9 true/false city statements [Marks and Tegmark, 2024] (layer 12). "Layer L" is the Hugging Face hidden state after L blocks. Controls: C1 smooth Gaussian processes and random walks over an ordinal index in $\mathbb{R}^{768}$ (expected: ringing); C2 matched-covariance Gaussians for each dataset (expected: blob); C3 the same inputs through randomly initialized weights.

Statistics. Percentile bootstrap CIs (2000 replicates); paired bootstrap and Wilcoxon signed-rank tests with Holm correction [Holm, 1979]; for real data, 200 random 80% subsamples (PCA recomputed) give label frequencies, a 5-fold held-out point residual compares method, B1, helix and ringing fits, and exact binomial CIs plus a label-permutation null test agreement with the literature.

6 Results

6.1 Synthetic test set

Table 2 compares all systems. On the full test set ($n = 39000$ clouds), the method selects the correct family for 87.9% [87.6, 88.2] of the clouds, with a median Chamfer distance to the noise-free ground-truth shape of 0.156 [0.154, 0.158] (in units of the RMS radius of the cloud). On the baseline subset, where every system sees the same $n = 3120$ clouds, the method reaches 87.4%, against 64.5% for the feature classifier B2, 52.2% for the classical fit-every-family pipeline B1, and 7.7%–7.9% for the trivial baselines.

Preregistered hypotheses (Table 3). H1 is supported: the paired macro-F1 difference between the method and B1 is 0.374 [0.358, 0.390]. H2 is supported: the paired difference in median Chamfer distance is -0.055 [-0.065, -0.046]. Refinement (H5b) lowers the median Chamfer distance by a small but significant amount, -0.003 [-0.005, -0.002]; because the family is selected by the classifier before refinement, refinement cannot change accuracy (A3 has the same accuracy as the method by construction). H5a (Chamfer versus parameter-space loss) requires the ablation runs, which are postponed (Sec. 7).

Table 3: Preregistered hypotheses. p-values are Holm-corrected over the synthetic comparisons; H3 uses an exact binomial CI and a 10,000-permutation null over predicted coarse labels.

Hypothesis	Comparison	Estimate [95% CI]	p (Holm)	n	Verdict
H1	macro-F1(method) - macro-F1(B1)	0.3741 [0.3579, 0.3903]	< 1e-16	3120	supported
H2	median Chamfer(method) - median Chamfer(B1)	-0.0554 [-0.0645, -0.0462]	< 1e-16	3120	supported
H5b_refinement	median Chamfer(method) - median Chamfer(no refinement)	-0.0033 [-0.0046, -0.0021]	< 1e-16	39000	supported
H3	Coarse-label agreement with literature on real datasets	7/12 = 58% [28, 85] (exact binomial)	perm. p = 0.125	12	not supported
H4	Controls: matched Gaussians → blob; ordinal processes → ringing (not helix)	blob rate 86%; ringing 60% vs helix 40%	-	-	supported

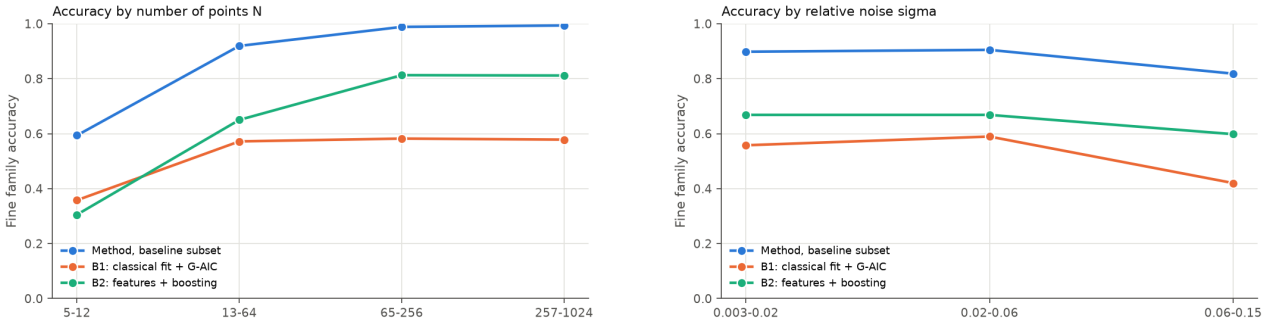


Figure 1: Fine accuracy by number of points and by relative noise on the baseline subset (the same clouds for every line). A3 is omitted because its accuracy equals the method's by construction.

Where the gap comes from. Selecting among the network's own refined candidates with geometric AIC (A5) lowers accuracy to 64.6%, close to B2. Model selection, not fitting, is what separates the systems: with nested families and realistic noise a residual-versus-complexity criterion often prefers the more flexible family, while the classifier has learned the prior over families and noise that generated the data. The advantage of the method is largest for the smallest clouds (Fig. 1). The hardest families are cone, helix and paraboloid; most of their errors are confusions with each other (Appendix, Fig. 6). Temperature scaling with $T = 0.772$ fitted on the validation stream lowers the expected calibration error from 0.044 to 0.022 and the NLL from 0.397 to 0.373 (Appendix, Fig. 6).

6.2 Real activations

Table 4 lists every real dataset and control, and Fig. 3 shows the clouds with the fitted shapes. The coarse label agrees with the literature on 7/12 preregistered datasets; the label-permutation null gives $p = 0.125$, so **H3 is not supported** (it is not falsified either, since agreement is not below 50%).

Successes. The toy model of superposition with 6 features (R3) is labelled polytope ($p = 1.00$), with a fitted vertex on every feature direction (the octahedron of Elhage et al., 2022). Llama's character-count manifold (R4) is labelled ringing ($p = 0.74$), a curve as in Gurnee et al. [2025]; under the preregistered rule a helix claim is not made because ringing is not rejected. Temperature (R8a) is labelled ringing, the arch PCA produces for a smoothly varying ordinal variable [Novembre and Stephens, 2008, Antognini and Sohl-Dickstein, 2018], which agrees with the "line" of Bhalla et al. [2026] at the coarse level. Colours (R7) are labelled cone, a curved surface as reported (paraboloid in Bhalla et al., 2026).

Failures. (i) GPT-2's positional embeddings (R1) are a 1D spiral whose radius varies along the curve; no family represents it (helix has constant radius, ringing three harmonics), and the classifier selects sphere ($p = 0.65$), a surface the spiral lies on. The held-out residual cannot reveal this error because held-out points also lie on the sphere. (ii) Weekdays and months (R5) form one tight cluster per token, 7 and 12 clusters, more than the 6 centroids the clusters family allows; the fitter threads a helix through them. (iii) Years (R6) and the grokked embeddings (R2) keep only 25% and 32% of the variance in the top three components and are flagged ($\lambda_3/\lambda_4 < 1.5$): the 3D view cuts through a higher-dimensional structure, as expected for the several Fourier circles of a grokked model [Nanda et al., 2023]. (iv) Age (R8b) is a branching curve and true/false city statements (R9) two overlapping clouds of very different spread; both are outside the vocabulary (labelled clusters and ringing).

Controls (H4). Matched-covariance Gaussians are labelled blob in 86% of draws and the ordinal-process controls ringing in 60% versus helix in 40%, so H4 passes as preregistered. A post-hoc split shows that this pass relies on the random walks (ringing in 100%): the smooth Gaussian-process controls alone are labelled helix in 80%. The classifier therefore confuses smooth ordinal processes with helices, the confound the preregistered helix rule guards against; for the fully evaluated Gaussian-process control the held-out residual favours ringing and the rule rejects the helix claim. Randomly initialised GPT-2 positional embeddings and grokking embeddings are

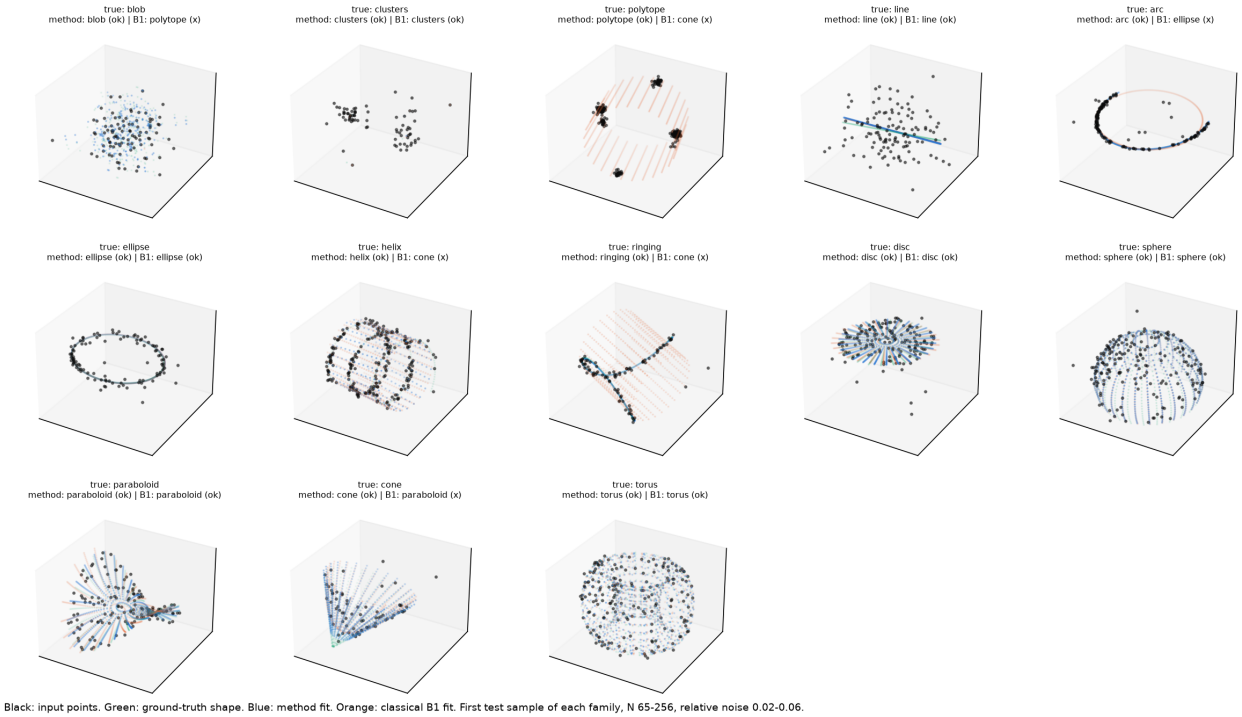


Figure 2: The first synthetic test cloud of every family (65–256 points, relative noise 0.02–0.06): input points (black), ground-truth shape (green), method fit (blue), classical B1 fit (orange).

labelled blob, but randomly initialised Llama is labelled clusters for the month prompts: Fig. 4 shows genuine clusters, one per input token, because a random transformer still separates token identities. The preregistered expectation "blob" for this control was wrong, not the fitter.

7 Discussion and limitations

What the results support. On synthetic clouds drawn from the 13-family generative model, a network trained only on synthetic data identifies the family and recovers the shape better than a classical pipeline that fits every family and selects with geometric AIC (H1, H2), and its probabilities are well calibrated after temperature scaling. This is a *systematic* claim within the synthetic distribution (39000 test clouds over all point-count and noise bins), with paired bootstrap CIs. The ablation A5 localises the advantage to model selection: given the same refined candidates, geometric AIC selects much worse than the learned classifier.

What the results do not support. The fitter is not yet a reliable replacement for by-eye shape calls on real activations. Coarse agreement with the literature (H3) is not above a permutation null, and the failures have identifiable causes: curves outside the vocabulary (GPT-2 positions), more than six discrete groups (weekdays, months), clusters of unequal spread (true/false statements), branching curves (age), and top-3 projections of higher-dimensional structure (years, grokked embeddings). In each case the classifier still returns a confident label, because a closed-set classifier must pick one of its 13 families. Smooth ordinal controls are frequently labelled helix (80% of the Gaussian-process controls, post hoc), so a helix label from the classifier alone is not evidence of a helix; the preregistered two-part rule (probability ratio and held-out residual) is necessary and rejected the helix claim where it was evaluated.

Implications for practice. (1) Report the shape of the top-3 projection together with the explained variance and the λ_3/λ_4 gap; a low gap means the 3D picture is a slice. (2) Treat "helix" as a hypothesis to be tested against the PCA ringing artifact [Novembre and Stephens, 2008, Antognini and Sohl-Dickstein, 2018, Shinn, 2023] with held-out data, not as a reading of the plot. (3) Controls with random weights are not structureless

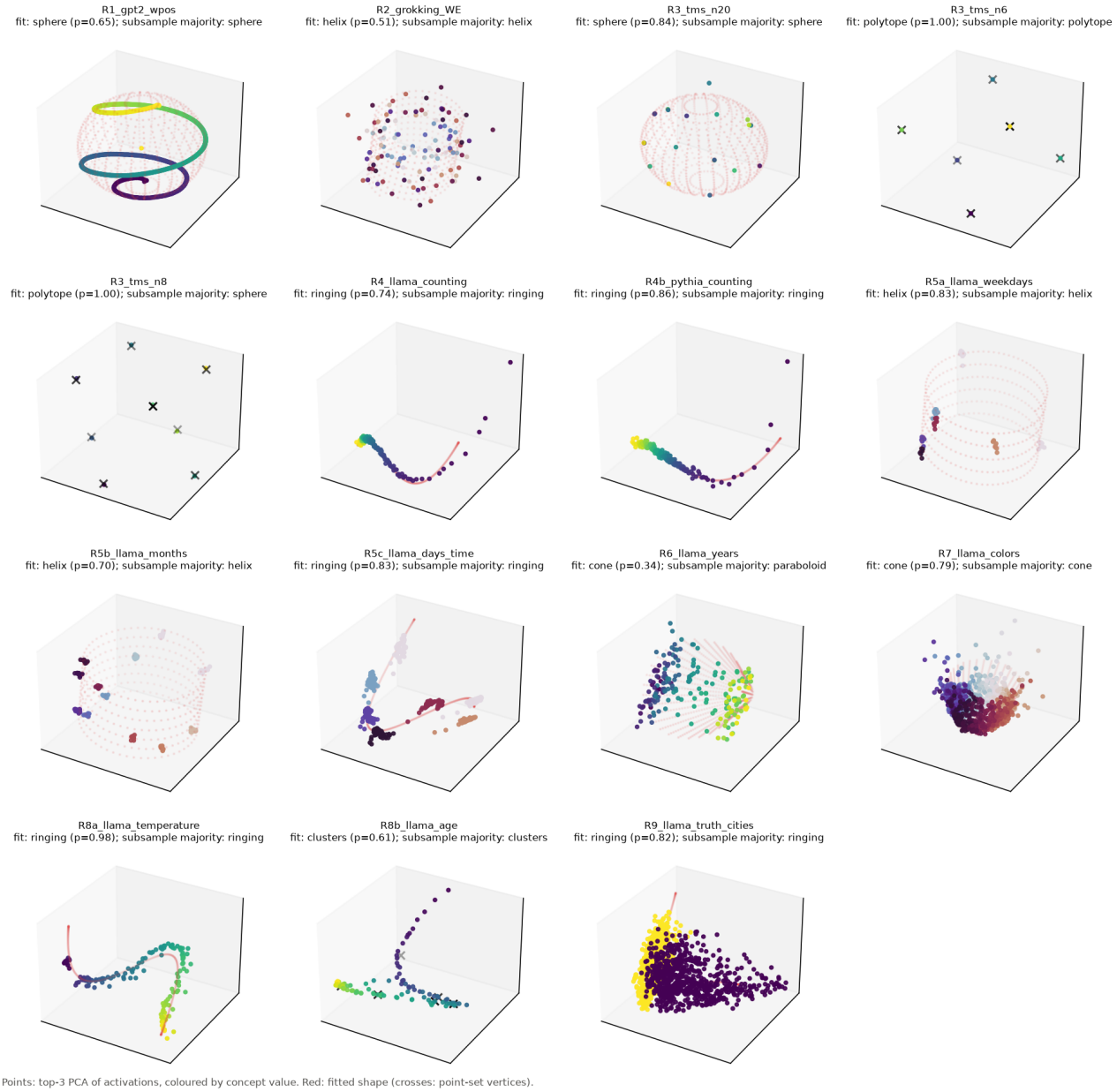


Figure 3: Real activation clouds in their top-3 PCA coordinates, coloured by concept value, with the fitted shape (red; crosses for point-set vertices). Titles give the full-cloud fit with its calibrated probability and the majority label over 200 random 80% subsamples.

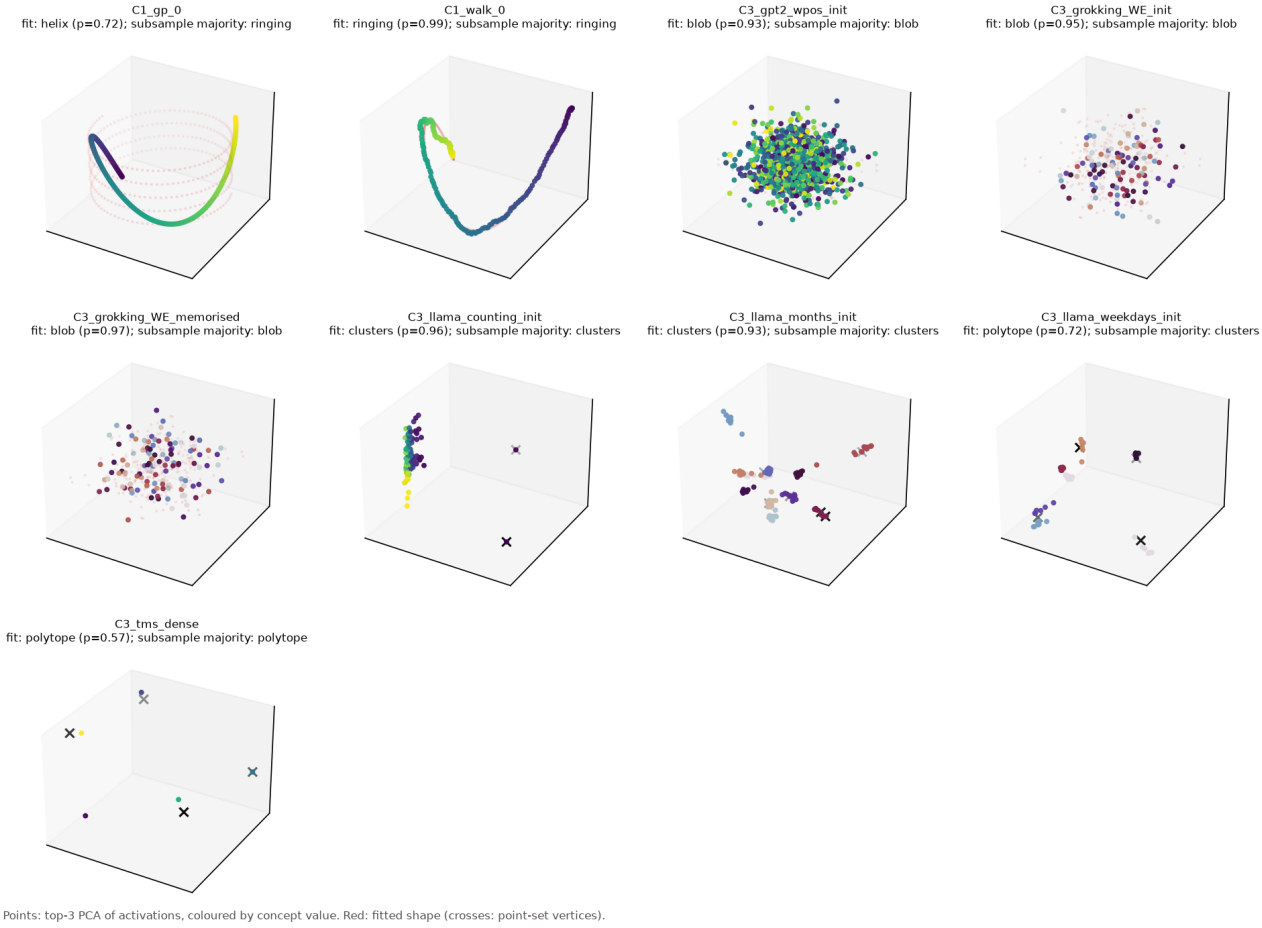


Figure 4: Controls: a smooth Gaussian process and a random walk over an ordinal index (C1), and randomly initialised or untrained networks (C3).

Table 4: Real datasets and controls (C1 shows the first of 10 draws of each process). Flags: $\lambda_3/\lambda_4 < 1.5$ (gap); OOD: residual above the synthetic 95th percentile of correct fits at that N. Held-out residuals, matched-Gaussian rates and model/layer details are in the full table in `reports/exp02_real_activations.md`.

Dataset	N x D	Top-3 var.	Expected (coarse / fine)	Fit (full cloud)	p	Subsample majority (freq.)	Flags
C1_gp_0 (prereg)	200 x 768	99%	1d / ringing	helix	0.72	ringing (58%)	-
C1_walk_0 (prereg)	200 x 768	83%	1d / ringing	ringing	0.99	ringing (100%)	-
C3_gpt2_wpos_init (prereg)	1023 x 768	1%	3d / blob	blob	0.93	blob (100%)	gap<1.5
C3_grokking_WE_init (prereg)	113 x 128	9%	3d / blob	blob	0.95	blob (100%)	gap<1.5
C3_grokking_WE_memorised	113 x 128	15%	- / -	blob	0.97	blob (100%)	gap<1.5
C3_llama_counting_init (prereg)	150 x 4096	14%	3d / blob	clusters	0.96	clusters (98%)	gap<1.5, OOD
C3_llama_months_init (prereg)	144 x 4096	19%	3d / blob	clusters	0.93	clusters (82%)	gap<1.5, OOD
C3_llama_weekdays_init (prereg)	49 x 4096	34%	3d / blob	polytope	0.72	clusters (70%)	gap<1.5, OOD
C3_tms_dense	5 x 3	100%	- / -	polytope	0.57	polytope (100%)	-
R1_gpt2_wpos (prereg)	1023 x 768	91%	1d / helix	sphere	0.65	sphere (74%)	-
R2_grokking_WE (prereg)	113 x 128	32%	1d / arc, ellipse	helix	0.51	helix (43%)	gap<1.5
R3_tms_n20	20 x 3	100%	0d / polytope, clusters	sphere	0.84	sphere (78%)	-
R3_tms_n6 (prereg)	6 x 3	100%	0d / polytope, clusters	polytope	1.00	polytope (100%)	-
R3_tms_n8 (prereg)	8 x 3	100%	0d / polytope, clusters	polytope	1.00	sphere (56%)	-
R4_llama_counting (prereg)	150 x 4096	69%	1d / helix	ringing	0.74	ringing (92%)	-
R4b_pythia_counting	150 x 1024	75%	1d / helix	ringing	0.86	ringing (100%)	-
R5a_llama_weekdays (prereg)	49 x 4096	66%	1d / arc	helix	0.83	helix (58%)	gap<1.5
R5a_llama_weekdays_means (prereg)	7 x 4096	78%	1d / arc	sphere	0.40	helix (74%)	gap<1.5
R5b_llama_months (prereg)	144 x 4096	55%	1d / arc	helix	0.70	helix (87%)	gap<1.5
R5b_llama_months_means (prereg)	12 x 4096	60%	1d / arc	helix	0.40	helix (51%)	gap<1.5
R5c_llama_days_time	336 x 4096	58%	1d / arc	ringing	0.83	ringing (86%)	gap<1.5
R6_llama_years (prereg)	199 x 4096	25%	1d / helix	cone	0.34	paraboloid (62%)	gap<1.5
R7_llama_colors (prereg)	900 x 4096	66%	2d / paraboloid	cone	0.79	cone (83%)	gap<1.5
R8a_llama_temperature (prereg)	150 x 4096	90%	1d / line	ringing	0.98	ringing (100%)	-
R8b_llama_age (prereg)	99 x 4096	85%	1d / line	clusters	0.61	clusters (96%)	OOD
R9_llama_truth_cities (prereg)	1496 x 4096	53%	0d / clusters	ringing	0.82	ringing (100%)	OOD

when inputs differ in token identity; a token-identity-matched control is the right null for prompt-based datasets. (4) Exclude sequence position 0 (BOS or the first token) when averaging activations: attention-sink tokens carry massive activations [Sun et al., 2024, Xiao et al., 2024] that can dominate class means.

Limitations.

- **One seed, no trained ablations yet.** All results come from one trained model (`main_s0`); the seed, learning-rate and inference-hyperparameter sweeps and the trained ablations A1, A2, A4, A6 (including H5a) are implemented but postponed to later stages (`reports/deviations.md`). Synthetic CIs therefore reflect test-sample variation only, not training variation.
- **Baselines on a subset.** B1 and B2 were run on 20 clouds per family and cell (3120 clouds) because B1 refines 61 candidate fits per cloud; all paired comparisons use this subset.
- **Closed vocabulary.** 13 families with label margins between nested families; shapes outside the vocabulary are forced into one of them. There is no abstention beyond the OOD and gap flags.
- **Synthetic prior.** Training data follow hand-designed families and nuisances; real activation clouds differ (non-uniform density along curves, token-level clusters, outliers), which is the main source of the synthetic-to-real gap.
- **Top-3 PCA only.** All statements concern the top-3 projection, as the task specifies; intrinsic manifolds can differ, and causal roles of these representations are not tested.
- **Literature labels as ground truth.** H3 compares against published descriptions that were themselves made by eye, often with different models, layers or prompts (we use Llama-3.1-8B for all LLM datasets).

8 Conclusion

We built a catalog of empirically reported neural geometries, a 13-family shape vocabulary with unique canonical parameters, and a Set Transformer that maps a top-3 PCA point cloud to a family and its parameters, trained only on synthetic shapes with a shape-to-shape Chamfer loss. On held-out synthetic data it identifies the family in 87.9% of clouds and beats a classical fit-every-family pipeline with geometric-AIC selection on both family identification and shape recovery (H1, H2 supported). On real activations, coarse agreement with published shape descriptions (7/12) is not above a permutation null (H3 not supported), and controls show that smooth ordinal processes are often mistaken for helices. The concrete failure modes—out-of-vocabulary curves, many discrete groups, unequal clusters and low-variance projections—point to the next steps: a coverage check that penalises fitted shapes without supporting points, a more flexible Fourier-curve family, more cluster centroids, and abstention for clouds outside the training distribution.

Reproducibility

All code, configurations, logs, the preregistration and the deviation log are at https://github.com/BurnyCoder/geometrical-shape-fitting-in-llm-mechanistic-interpretability. One command (`uv run python run.py report --config main.yaml`) regenerates every number, table and figure in this paper from the results files. The trained fitter is on the Hugging Face Hub (collection "Fitting geometric shapes to neural geometry"), and an interactive explorer of all fits is at https://burnycoder.github.io/geometrical-shape-fitting-in-llm-mechanistic-interpretability/.

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A Catalog of empirically reported shapes

Shape	Concept	Model / layer	Top-3 PCA	Family	Source
circle (overall a cone; PC1 = intensity/radius)	days of the week, months of the year (also 20th-century years in GPT-2)	GPT-2-small L7 (SAE clusters); Mistral-7B L8/L30; Llama-3-8B L5	yes (PCs 1-2 or 2-3)	arc	Engels et al. 2024/2025, Not All Language Model Features Are
circle ("days of the week organized in a cycle")	days of the week (prompt "It's time on day")	Llama-3.1-8B L19, last token	yes	arc	Bhalla et al. 2026, Do Sparse Autoencoders Capture Concept M
circles (one per key Fourier frequency, "clock")	residues in modular addition (a+b mod p)	1-layer transformer trained from scratch, p=113 / p=59	yes (PCs 1-2; PC3 belongs to another frequency)	arc	Nanda et al. 2023, Progress measures for grokking via mechan
ring / square grid / hexagonal lattice (in-context graph structure)	graph defined in context by random walks over tokens	Llama-3.1-8B L26	yes (top-2 PCs)	arc	C. F. Park et al. 2025, ICLR: In-Context Learning of Represe
circle (valence-arousal circumplex)	emotions	Llama-3.1-8B, Qwen3-8B/14B	yes (2D)	arc	Sun et al. 2026, Valence-Arousal Subspace in LLMs: Circular
ring	head direction (neuroscience analogue)	mouse anterodorsal thalamic nucleus	yes	arc	Chaudhuri, Gercek, Pandey, Peyrache, Fiete 2019, The intrins
helix	learned absolute positional embeddings (W_pos, 1024 x 768)	GPT-2 small/medium/large (weak in XL)	yes (90% variance in 3 PCs for GPT-2 small)	helix	Yedidia 2023, GPT-2's positional embedding matrix is a helix
twisting helix-like rippled curve (1D manifold in 6D)	character count since last line break	Claude 3.5 Haiku L2; replicated in Llama-3.1-8B L5 / Qwen3-8B L5 (PC1-3 = 97% variance)	yes	helix	Gurnee et al. 2025, When Models Manipulate Manifolds: The Ge
generalized helix (line + circles with periods 2, 5, 10, 100)	integers 0-360	GPT-J, Pythia-6.9B, Llama-3.1-8B (residual after layer 0)	partial (PC1 linear; helix only in fitted subspace)	helix	Kantamneni & Tegmark 2025, Language Models Use Trigonometry
helix	years (prompt "The date is year")	Llama-3.1-8B L19, last token	unknown (not plotted)	helix	Bhalla et al. 2026, Do Sparse Autoencoders Capture Concept M
periodic Fourier features (circles per digit/period)	numbers in addition	pre-trained LLMs (GPT-2 family and others)	unknown	arc	Zhou et al. 2024, Pretrained Large Language Models Use Four
line	temperature ("Today it's f degrees Fahrenheit outside"), age ("They are age years old."), formality, sentence length	Llama-3.1-8B L19, last token	yes (Fig. 2)	line	Bhalla et al. 2026, Do Sparse Autoencoders Capture Concept M
line / linear map (latitude, longitude, year)	places and historical entities	Llama-2 7B/13B/70B, mid-depth	no (top PCs separate entity types)	line	Gurnee & Tegmark 2024, Language Models Represent Space and T
sphere (cities); circle (dates, time of day); semicircle (events 1900-2000); log-line (durations); clusters (seasons)	time and space concepts	Llama-3.x, Qwen2.5, Gemma-2	no (paper: PCA misses subspaces not aligned with max variance)	sphere	Tiblias et al. 2025 (TMLR 2026), Hypothesis-Driven Feature M
paraboloid ("circular hue and lightness dimensions")	colors (prompt "The hex code h_code is for the color")	Llama-3.1-8B L19, last token	yes (Fig. 2)	paraboloid	Bhalla et al. 2026, Do Sparse Autoencoders Capture Concept M
disc ("pizza") or torus in first layer; circle in later layers	modular addition	small MLPs and transformers trained from scratch	disc yes; torus needs 4D	torus	Moisescu-Pareja et al. 2025, On the geometry and topology of
filled disc ("pizza") of embeddings	residues in modular addition (pizza algorithm, attention-rate-dependent)	1-layer transformers trained from scratch	yes (PCs 1-2)	disc	Zhong, Liu, Tegmark, Andreas 2023, The Clock and the Pizza:
torus	grid-cell modules (neuroscience analogue)	rat medial entorhinal cortex	no (needs 4D)	torus	Gardner et al. 2022, Toroidal topology of population activit
narrow cone (anisotropy)	all contextual embeddings	GPT-2, BERT, ELMo	partial (PC1 dominance)	cone	Ethayarajh 2019, How Contextual are Contextualized Word Repr
two linearly separated clusters	true vs false statements	LLaMA-13B (and LLaMA-2 family)	yes	clusters	Marks & Tegmark 2024, The Geometry of Truth
antipodal pairs, triangles, pentagons, tetrahedra, square antiprisms	features in superposition	toy ReLU autoencoders	n/a (weights are already low-dimensional)	polytope	Elhage et al. 2022, Toy Models of Superposition
simplices / polytopes; hierarchy orthogonal	WordNet categories	Gemma, LLaMA-3	no	polytope	K. Park, Choe, Jiang, Veitch 2024, The Geometry of Categoric
parallelogram "crystals", lobes	SAE decoder directions	(SAE features of LLMs)	no	polytope	Li, Michaud, Baek, Engels, Sun, Tegmark 2025, The Geometry o
hierarchical tree	geographical coordinates → country	Llama-3.1-8B L19	unknown	out-of-vocab	Bhalla et al. 2026
fractal inside a 2-simplex	belief states of a hidden	4-layer transformer, d=64	no	out-of-	Shai et al. 2024, Trans-

B Additional results

B.1 Unambiguous synthetic subset

Synthetic clouds that a simpler family also fits equally well (judged from B1's per-family fits) are tagged ambiguous (34.0% of the baseline subset). Table 5 repeats Table 2 without them.

Table 5: Systems on the baseline subset without ambiguous clouds.

System	n	Fine accuracy [95% CI]	Macro-F1 [95% CI]	Coarse accuracy	Superset-OK accuracy	Median Chamfer to GT [95% CI]	Mean residual	Median axis error (deg)
Method (network + refinement)	2060	94.0% [93.1, 95.0]	0.922 [0.909, 0.935]	96.4%	94.3%	0.119 [0.111, 0.128]	0.091	0.7
A3: no refinement	2060	94.0% [93.1, 95.0]	0.922 [0.909, 0.935]	96.4%	94.3%	0.127 [0.119, 0.132]	0.101	1.1
A5: G-AIC selection	2060	72.0% [70.3, 73.8]	0.654 [0.639, 0.669]	81.5%	76.5%	0.138 [0.134, 0.144]	0.097	0.6
A5: hybrid selection	2060	92.6% [91.4, 93.7]	0.902 [0.887, 0.916]	94.8%	93.1%	0.119 [0.111, 0.128]	0.090	0.7
B1: classical fit + G-AIC	2060	69.7% [67.6, 71.6]	0.636 [0.615, 0.652]	82.5%	75.4%	0.154 [0.148, 0.160]	0.103	0.6
B2: features + boosting	2060	72.3% [70.2, 74.3]	0.697 [0.674, 0.716]	92.5%	76.2%	0.152 [0.147, 0.159]	0.099	0.6
B3a: always blob	2060	4.5% [3.6, 5.3]	0.007 [0.005, 0.008]	4.5%	4.5%	0.523 [0.516, 0.529]	0.119	10.2
B3b: always circle	2060	7.3% [6.4, 8.5]	0.011 [0.009, 0.012]	36.3%	7.3%	0.528 [0.516, 0.538]	0.273	0.3
B3c: random shape	2060	8.3% [7.2, 9.6]	0.079 [0.068, 0.091]	34.1%	11.6%	1.114 [1.094, 1.138]	0.582	58.6

B.2 Training, confusion and calibration

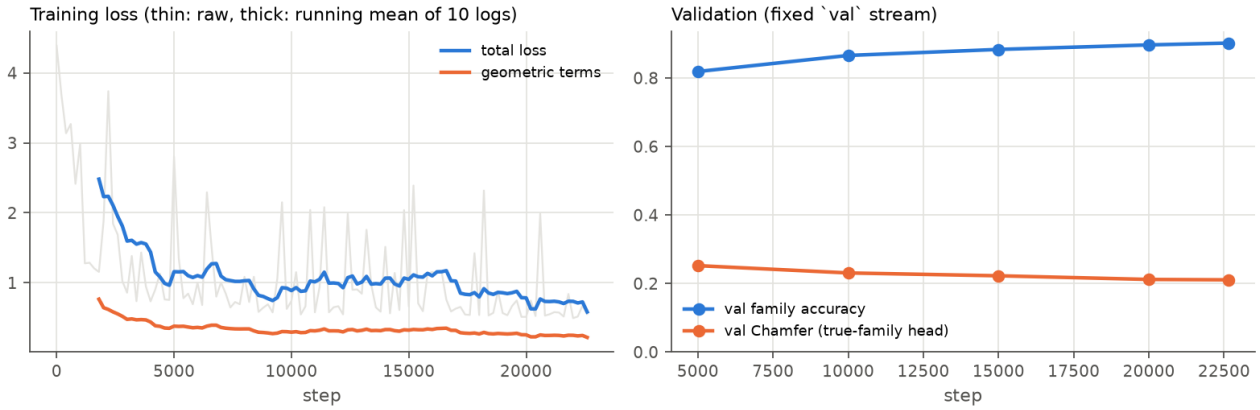


Figure 5: Training loss (left) and validation family accuracy and teacher-forced Chamfer distance on the fixed validation stream (right). Steps 0–9999 of the loss curve were recovered from the log of the crashed first attempt (`reports/deviations.md`).

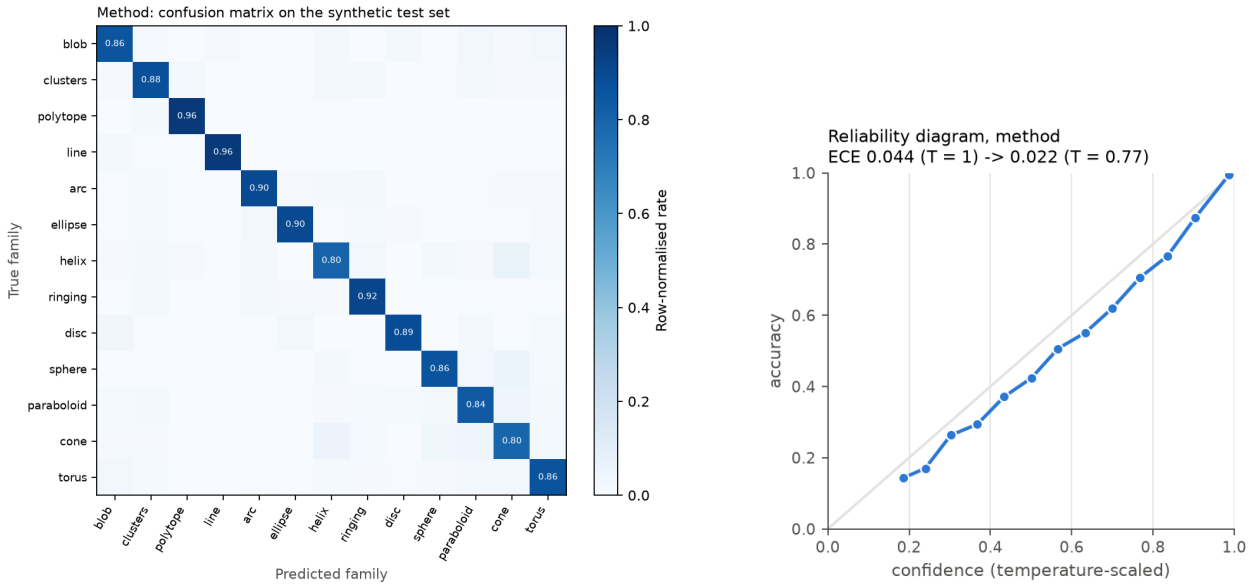


Figure 6: Left: confusion matrix of the method on the full synthetic test set (row-normalised). Right: reliability diagram after temperature scaling.

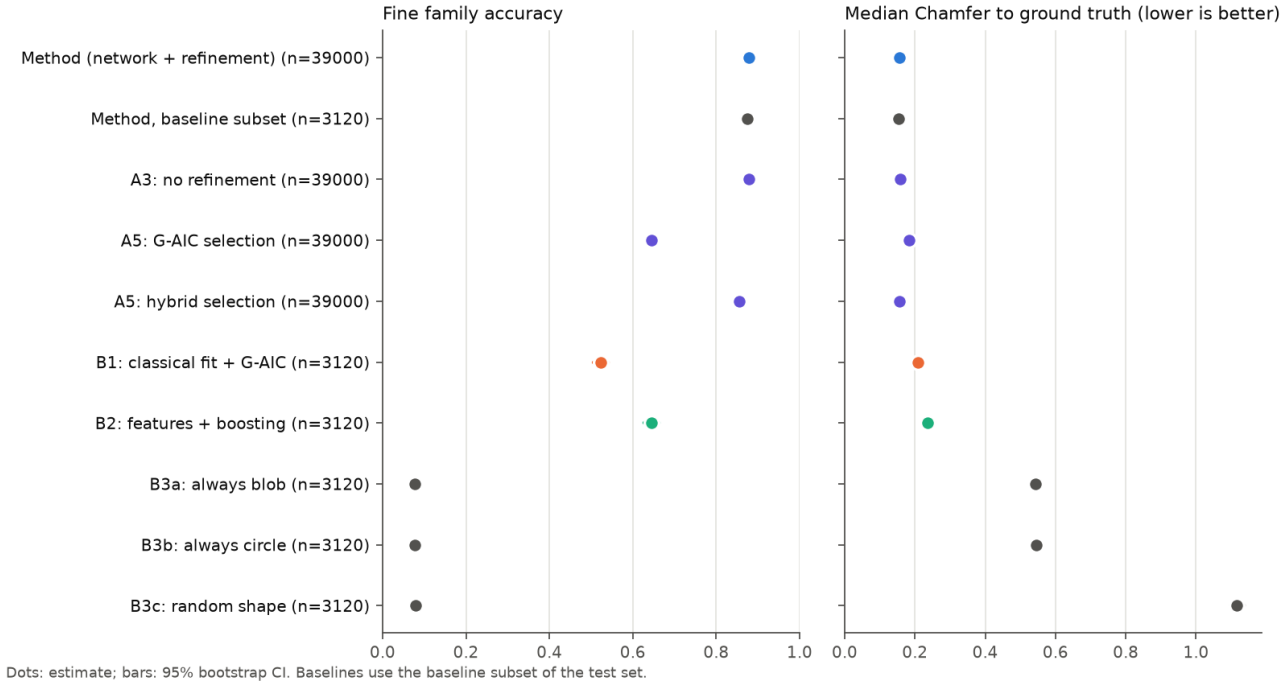


Figure 7: Fine accuracy and median Chamfer distance of all systems with 95% bootstrap CIs.

B.3 Models

Table 6: Trained fitters.

Run	Encoder	Training objective	Data / curriculum	Params	Steps	Minutes	Val. accuracy	Test fine accuracy	Test median Chamfer
main_s0	Set Transformer	CE + Chamfer + residual + aux	full synthetic mix	1,712,093	22643	110	90.2%	87.9%	0.156

AI statement

This codebase and paper were written by Claude Code (Anthropic), an AI coding agent, which executed a specification provided by the human author; the human author provided the research question, constraints and standards to follow. Human cleanup and rewriting are in progress.